MPOWERED Architecture Guide

Overview

MPOWERED is a React Native app built with Expo 57 and TypeScript. The same codebase supports iOS, Android, and web. Expo Router handles navigation, React Context holds shared state, and Supabase provides authentication and a PostgreSQL database.

Most database work follows this path:

```
Route -> Feature screen/context -> Service -> Supabase
```

There is no separate application server. Database calls live in `src/services/`, while sign-in and session handling also call Supabase directly from screens and providers.

Project layout

Paths below are relative to `musculoskeletal-mpowered/`.

```
src/
|-- app/                 Routes and navigation layouts
|-- features/            Screens, state, and logic for each feature
|-- components/          Shared UI, navigation, typography, and motion
|-- services/            Database calls and native/web adapters
|-- utils/validation.ts  Phone validation and formatting
`-- theme.ts             Shared colours and design tokens
assets/                  Images, fonts, and font licences
supabase/
|-- config.toml          Local backend configuration
`-- migrations/          Database schema and access policies
tests/                   TypeScript tests and a separate SQL RLS test
app.json                 Expo targets, plugins, and permissions
app.config.js            GitHub Pages base-path setting
package.json             Dependencies and scripts
```

Features include onboarding, Home, the five assessment areas, My Health, reflections, Care Planner, and settings. Each feature keeps its screens, context, types, and data helpers together. A route usually selects a feature screen and supplies navigation callbacks. New database operations belong in the relevant service rather than in screen layout code.

Navigation and state

`src/app/_layout.tsx` loads fonts and runs `SessionLoader` before showing the app. It wraps the navigator in providers for onboarding, Home progress, reflections, My Health, Care Planner, settings, and motion.

The five assessment groups each have a nested layout and their own context for unfinished answers. The root providers hold data needed across sections, such as saved assessments and appointment plans.

The main destinations are Pain Tracker (`/home`), My Health, Care Planner, and Settings. They use bottom navigation on smaller screens and a sidebar from 1024 logical pixels. Splash and onboarding have neither.